BHAVESH GURNANI

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EDUCATION

- **Indian Institute of Technology Delhi (IIT Delhi)** *May 2026*
  - B.Tech and M.Tech in Computer Science & Engineering (Dual Degree) | CGPA: 8.52/10.0
- **Lord Buddha Public School, Kota** *Apr 2021 (CBSE XII), Apr 2019 (CBSE X)*
  - Class XII (PCM): 84% | Class X: 94%

SCHOLASTIC ACHIEVEMENTS

- **JEE Advanced 2021:** All India Rank **116** (out of approx. 200,000 candidates)
- **JEE Mains 2021:** All India Rank **493** (out of >1,000,000 candidates); **100% Percentile** in Mathematics
- **KVPY Fellowship 2020 (IISc):** All India Rank **305** (SA Stream)
- **NTSE Scholar 2019 (NCERT):** Ranked among Top **2000** students nationwide
- **INMO Qualifier:** Ranked among Top **300** students nationwide

EXPERIENCE

- **Quantitative Researcher** | *Ebullient Securities* | *Gurugram* *July 2026 – Present*
  - Researching alpha signals and systematic trading strategies for US options markets.
- **Research Intern** | *Microsoft Research* *Feb 2026 – April 2026*
  - Built conversational agents for medical diagnosis using multi-turn dialogue modeling and clinical reasoning.
- **AI Security Intern** | *National University of Singapore — Prof. Prateek Saxena, KISP Lab* *Dec 2025 – April 2026*
  - Extended *OverThink* (Kumar et al., 2025; arXiv:2502.02542) with a filtering-evasive slowdown attack on LLMs.
  - Showed it increases reasoning-token generation without changing outputs, revealing compute-amplification risks.
- **Post-Training Research Intern** | *Neosigma* | *Remote* *Dec 2025 – Jan 2026*
  - Improved 128k-context agentic LLM post-training by tool-call truncation during SFT, improving context retention.
  - Set up AWS multi-node training infrastructure for scalable SFT, DPO, and GRPO with `verl` at 128k context.
- **Quantitative Researcher Intern** | *Ebullient Securities* | *Gurugram* *May 2025 – July 2025*
  - Built and backtested alpha signals for short-term equity directionality from equities data.
  - Designed and live-tested a custom trading strategy in C++ using top alphas (40% annualised, Calmar 12).
  - Automated alpha-generation pipeline using LLMs with custom evaluation loops for iterative refinement.
- **Software/Research Engineer Intern** | *CompilerAI Labs Private Limited* | *New Delhi* *May 2024 – July 2024*
  Project under the guidance of Prof. Sorav Bansal
  - Implemented MISRA C static analysis rules within Clang compiler across Preprocessor, AST, and LLVM stages.
  - Enhanced Equivalence Checker tool for formal verification of compiled executable code against C source code.

TEACHING ASSISTANTSHIP

- **TA, COL772: Natural Language Processing** | *IIT Delhi* | *Prof. Mausam* *Jan 2026 – May 2026*
  - Designed assignments and quizzes; built autograder to evaluate student submissions; resolved Piazza queries.
- **TA, COL333: Artificial Intelligence** | *IIT Delhi* | *Prof. Mausam* *Jul 2025 – Dec 2025*
  - Created quizzes and assignments; evaluated student scripts.

PUBLICATIONS

- **ProbMedTOD: A Probabilistic Task-Oriented Dialogue System for Patient History Taking**
  *Vishal Vivek Saley, **Bhavesh Gurnani**, Dinesh Raghu, Mausam* *NeurIPS 2026*
  - Introduced **ProbMedTOD**, which auto-synthesizes clinical Bayesian Networks from a few clinical notes.
  - Integrated BayesNets with SLMs: a BayesNet Agent tracks diagnosis posteriors, a Policy Agent picks questions.
  - Outperformed LLM-only and RAG baselines on DDXPlus and MIMIC-IV (21-pt MRR gain), esp. for smaller models.
- **Hierarchical Masked Diffusion Language Models**
  *Parul Gupta, Avni Aggarwal, **Bhavesh Gurnani**, Arnab Kumar Mondal, Dinesh Khandelwal, Dinesh Garg, Parag Singla* *In Review*
  - Proposed **HMDLM**, a two-stage diffusion model that predicts token-level generation priorities before decoding.
  - Derived novel ELBO with per-level noise schedules for controllable non-autoregressive generation.
  - Achieved 0.91 PPL improvement on LM1B, 1.4% gain on GSM8K, and 3% on code generation.

SELECTED PROJECTS

- **Hierarchical Masked Diffusion Language Models** | ***MTP under Prof. Parag Singla*** *June 2025 – May 2026*
  - Implemented hierarchical masked-diffusion with staged generation: descriptions → comments → code tokens.
  - Achieved empirical improvements on LLADA for MBPP and HumanEval code generation; basis for HMDLM paper.
- **Improving Dialogue Systems for Medical Diagnosis** | ***BTP under Prof. Mausam*** *July 2024 – May 2025*
  - Synthesized clinical BayesNets from notes via ontology building, structure prediction, and parameter estimation
  - Designed BayesNet inference for multi-turn diagnosis, beating RAG baselines; basis for ProbMedTOD (NeurIPS 2026)

COURSE PROJECTS

- **Object-Centric Learning for Scene Decomposition** | ***DL Course Final Project*** *Jan 2024 – Apr 2024*
  - Implemented Slot Attention on CLEVRTex and designed slot-conditioned diffusion model with custom U-Net
  - Benchmarked performance using ARI metric and achieved interpretable object-centric scene representations
- **Intelligent Bot for Alternating Markov Games** | ***AI Course Final Project*** *Aug 2023 – Dec 2023*
  - Developed AI agents for strategic games using depth-limited Minimax with domain heuristics optimized via RL
  - Implemented DQN enhanced with MCTS inspired by Expert Iteration, achieving superior gameplay performance
- **Lexer, Parser, Interpreter for Language with Rationals** | ***Prof. S. Arun Kumar*** *Jan 2023 – Apr 2023*
  - Built complete compiler with custom BigInt/Rational modules, lexer using ML-Lex, and parser with ML-Yacc
  - Implemented interpreter with static scoping supporting variables, conditionals, loops, and recursion
- **Custom MultiThreading Library** | ***Prof. Abhilash Jindal*** *Feb 2023 – Mar 2023*
  - Designed a user-level multithreading library in C with round-robin scheduling and synchronization primitives
  - Benchmarked against POSIX pthread, demonstrating efficient scheduling with reduced overhead
- **Table Cell Classification for Question Answering** | ***Prof. Mausam*** *Feb 2024 – Mar 2024*
  - Designed a model to identify answer cells in custom tables for complex natural language queries
  - Built a two-stage transformer architecture: column predictor followed by row predictor conditioned on query and predicted columns
  - Prompt-tuned Gemma, improving exact match accuracy from 43% (two-model pipeline) to 67%
- **Text-to-Math Program Synthesis** | ***Prof. Parag Singla*** *Jan 2024 – Mar 2024*
  - Developed Seq2Seq models to generate executable math formulas from text problem descriptions, with an interpreter for answer computation
  - Implemented Bi-LSTM encoder-decoder with GloVe embeddings and Bahdanau attention; achieved 67% exact match and 73% execution accuracy using beam search
  - Fine-tuned BERT as encoder, improving to 78% exact match and 81% execution accuracy
- **Lay Summarization of Biomedical Research Articles** | ***Prof. Mausam*** *Apr 2024 – Jun 2024*
  - Fine-tuned PLMs on HPC clusters to generate layperson summaries of biomedical research articles
  - Applied PEFT techniques (Adapters, LoRA) with mixed-precision training for efficient large-scale fine-tuning
  - Evaluated Flan-T5 and BioGPT on relevance, readability, and factual consistency metrics

EXTRACURRICULAR ACTIVITIES

- **Tower Research Limestone Data Challenge:** Achieved **Top 50** selection **twice** across offline sessions
- **Inter IIT Tech Meet 2024:** Led team on Adobe Research problem; built a detection and explainability pipeline for AI-generated images, identifying visual artifacts and providing reasoning for model predictions
- **ML in Astronomy Workshop (2024):** Conducted a session on ML–Astronomy integration; built LSTM-Attention models on light curves for object classification [Code]
- **ARIES Research Team (2024):** Conducted NLP sessions and contributed to multiple industry research projects
- **Technical Executive, Economics Club, CAIC (2022–23):** Developed a React.js website to manage club competitions and events
- **Technical Engineer, Infinity Hyperloop, CAIC (2022–23):** Designed a PyQt5 interface for pod control; implemented sensor processing and STM communication over CAN protocols
- **Academic Mentor (2023):** Mentored 150+ freshers for MTL100 (Calculus) via hybrid doubt sessions